

Assessment in COMP34111 — AI and Games

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Semester 2, 2025–26

There are two components which contribute to the final mark of COMP34111: the group project and the final exam. Each are worth 50%. The problem sets do not contribute to the final mark. They have been included to aid learning, and as preparation for the final exam. The proforma also does not contribute to the final mark.

Final Exam

The final exam is a two-hour exam. It is old-style. There is a sheet of questions which you answer in answer booklets. At the exam, you will need pens, and an approved calculator is highly recommended.

1 Project Mark

Approach mark

For the project, 25% is *approach* and 25% is for *performance*. *Approach* is about how the team set about creating the bot. So, a bot which does not win many games but is based on a very sound approach could get a high approach mark, even if it does not get a good performance mark. A good approach could mean: using ideas not taught in the course, but discovered from the literature, based on a very clever or novel idea, created through systematic experimentation, or through advanced software engineering. Approach mark is assessed through the presentation and the journal.

The performance mark is broken down to win performance — which is how many games it wins, and speed performance — which is how fast it is. Win performance is worth 3 times more than speed performance, because it is easy to make a fast player simply by using a player which makes no calculations to choose its move. So, of the 25% performance mark, 18.75% is for win performance, and 6.25% is for speed performance.

1.1 Feedback

Feedback will come with the presentation marks, journal marks, and performance marks. These will be found on Canvas.

1.2 Approach mark

The approach mark is assessed by the lecturers based on the presentation which is given during the last two weeks of the semester, and what is described on the group journal. We are interested in how the choices you made in designing and improving your bot were informed by the following.

Good understanding of the methods used: We don't want you to produce a course on what we taught you. We want to see evidence in what you implement and what decisions you make that you understand the methods you are using.

Use of the literature: Was the system you built and the choices you made informed by work found in established literature?

Use of evidence and experiments: Did you do careful testing or comparative tests to choose parameters, evaluate approaches, etc. Did you show us the outcomes of these?

Effective implementation: Did you optimize the representation, the tree search, use parallelization, or use some other optimizations?

The Journal: There is a separate page on this. In short, it should show what each individual did, how it functioned as a group, and what was submitted in the end. It is very important that you document what you have done on your journal, both individually and as a group.

1.3 Marking of late and non-working bots

We will follow the University policy, which is 10 marks per 24 hour late. If a bot is submitted on time but does not run or work without later intervention, it could be considered late unless it can be quickly rectified by the students. Otherwise, it is marked as below.

0%: No bot submitted, late or otherwise.

32%: Bot submitted on time, but will not run.

40%: A bot submitted on time, but could not *in principle* win any games because it always either fails to produce commands which satisfy the protocol, always times out, or fails to play any games at all gets a mark of 40%.

1.4 Marking of win-performance of bots:

50% – 90%: Any bot which could in principle win games (i.e. it plays correctly) gets a mark in this range. The mark is determined by performance against other bots. It is our intention to run an all-against-all tournament.

Extra credit: Up to 10% marks can be awarded for an exceptionally well-thought out or novel approach. If a bot is faster than bots of similar win performance, or is better than many bots in *both* win performance and speed performance, it can gain marks here.

1.5 Marking of speed-performance of bots:

This will be based not on absolute run time, but on run-time *rank*, where the fastest bot is ranked 1 and the slowest bot is ranked N where N is the number of bots in the tournament.

50% – 90%: Any bot which could in principle win games (i.e. it plays correctly) gets a mark in this range. The mark is in proportion to the rank of the speed. A bot with the median speed gets a mark of 70%; the fastest which can in principle win a game gets 90%; the slowest gets 50%.

Extra credit: Up to 10% can be added to the mark. Often the run-times range over two or three orders of magnitude, which is not captured in the rank. Extra credit can be given to bots which are much faster than those of similar rank or similar win performance.

As mentioned previously, win performance is three times more important than speed. In other words,

$$\text{Performance score} = \frac{3 \times \text{wins score} + \text{speed score}}{4}.$$

1.6 Individual marks

Sections 1.2, 1.3, 1.4, and 1.5 describe how marks are assigned to groups. There also need to be mechanism to assign marks to individuals. Obviously, the exam does that. During the presentation, lecturers have the option of assigning different approach marks to different students. We rarely do this, because it is a team effort, everyone in the group is sharing the same approach, unless there is an unexcused absence, or a very big discrepancy between one member and others. The Journal marks are given individually.

If during the project work, it transpires that the group is not functioning as a team, we urge you to let us know as soon as possible so we can try to fix it. This could be because one member of the group is not engaged, or one group member is not being given the opportunity to engage by other members of the group. It is much better to fix the problem as soon as it emerges, than to fix it after the project is over. However, a peer assessment mechanism is provided to request differential marks for different members of your group.

1.7 Peer assessment

The course unit leader (Mingfei) has a strong bias towards assigning marks equally to every member of the group. You are a team; you are not competing with each other for marks. So, if someone in your group is not contributing or is not engaged at all, or much less than others, **please let us know as soon as possible, so we can attempt to fix it during the project**. It is possible for such a person to get lower marks if all group members agree.

The peer assessment mechanism is just to fix cases where someone did little or no work. Their marks can go down.